

1. **What is an Artificial Neural Network (ANN)?**

ANN is a computer model inspired by the human brain. It learns patterns from data and makes decisions or predictions.

2. **Why are ANNs inspired by the human brain?**

Because the human brain learns from experience using neurons. ANN works similarly by connecting artificial neurons to process information and learn.

3. **What are the main components of a neuron?**

- Inputs
 - Weights
 - Bias
 - Activation Function
 - Output
-

4. **What is the role of weights and bias in ANN?**

- **Weights** decide the importance of input values.
 - **Bias** helps shift the output and improves learning accuracy.
-

5. **What is an activation function?**

It decides whether a neuron should activate or not based on the input value.

6. **Why do we need activation functions in neural networks?**

Without activation functions, the network cannot learn complex patterns. They help ANN solve non-linear problems.

7. **Differentiate between biological neuron and artificial neuron.**

Biological Neuron	Artificial Neuron
Found in human brain	Found in computer model
Uses electrochemical signals	Uses mathematical calculations
Learns naturally	Learns from data

8. **What is supervised learning?**

A learning method where the model learns using labeled data (input + correct output).

Example: Spam email detection.

9. **What is unsupervised learning?**

A learning method where the model finds patterns from unlabeled data.

Example: Customer grouping.

10. **What is reinforcement learning?**

Learning by trial and error using rewards and penalties.

Example: Robot learning to walk.

11. **What is a single-layer perceptron?**

It is the simplest neural network with only one layer of output neurons.

12. **What is a multilayer perceptron (MLP)?**

An ANN with multiple layers (input, hidden, output) used to solve complex problems.

13. **What are input, hidden, and output layers?**

- **Input Layer:** Receives data
 - **Hidden Layer:** Processes data
 - **Output Layer:** Gives final result
-

14. **Why are hidden layers important?**

They help the network learn complex relationships and patterns from data.

15. **What is deep learning?**

A type of machine learning that uses deep neural networks with many hidden layers.

16. **Differentiate between shallow and deep neural networks.**

Shallow Network	Deep Network
------------------------	---------------------

Few hidden layers	Many hidden layers
-------------------	--------------------

Simple learning	Complex learning
-----------------	------------------

Shallow Network Deep Network

Less accurate More powerful

17. What is the purpose of neurons in hidden layers?

They extract important features and patterns from input data.

18. What is a feedforward neural network?

A network where data moves only in one direction: input → hidden → output.

19. What is recurrent feedback in neural networks?

It means the output is fed back into the network to remember previous information.

Used in RNNs.

20. What are the limitations of a single-layer perceptron?

It cannot solve complex non-linear problems like XOR.

21. Explain the sigmoid activation function.

Sigmoid converts input values into output between 0 and 1.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Used mainly for probability prediction.

22. What are the advantages and disadvantages of sigmoid?

Advantages:

- Smooth output
- Gives probability values

Disadvantages:

- Slow training
 - Vanishing gradient problem
-

23. What is the ReLU activation function?

ReLU outputs the input directly if positive, otherwise gives 0.

$$f(x) = \max(0, x)$$

24. **Why is ReLU widely used?**

Because it is simple, fast, and helps deep networks train efficiently.

25. **What is the tanh activation function?**

Tanh converts values between -1 and +1.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

It works better than sigmoid in many cases because output is centered around zero.

26. **Differentiate between sigmoid and tanh.**

Sigmoid

Tanh

Output range: 0 to 1 Output range: -1 to 1

Not zero-centered Zero-centered

Slower learning Faster learning

27. **What is softmax activation?**

Softmax converts outputs into probability values whose total sum is 1.

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

28. **In which applications is softmax used?**

Used in **multi-class classification** problems like:

- Image classification
 - Digit recognition
 - Language processing
-

29. **What is the vanishing gradient problem?**

During training, gradients become very small, so the network learns very slowly.

30. **Which activation functions help reduce vanishing gradients?**

- ReLU
- Leaky ReLU
- ELU

31. What is training in ANN?

Training means teaching the neural network using data so it can make correct predictions.

32. What is an epoch?

One complete pass of the entire training dataset through the neural network.

33. What is batch size?

The number of training samples processed at one time before updating weights.

34. What is iteration in ANN training?

One update of network weights after processing one batch.

35. What is forward propagation?

The process where input data moves through the network to produce output.

36. What is backpropagation?

The process of sending errors backward through the network to update weights.

37. Why is backpropagation important?

It helps the network learn by reducing errors and improving accuracy.

38. Explain gradient descent.

Gradient descent is an optimization method used to reduce error by updating weights step by step.

39. What is learning rate?

Learning rate decides how much weights change during training.

40. What happens if the learning rate is too high or too low?

- Too high → Training becomes unstable
 - Too low → Training becomes very slow
-

41. What is loss function?

A function that measures how much error the model has made.

42. Differentiate between loss function and cost function.

Loss Function

Error for one training example

Used for single sample

Cost Function

Average error for all examples

Used for whole dataset

43. What is Mean Squared Error (MSE)?

MSE calculates the average squared difference between actual and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

44. What is cross-entropy loss?

A loss function mainly used in classification problems to measure prediction accuracy.

45. What is optimization in ANN?

Optimization means improving the network performance by reducing loss.

46. Name different optimization algorithms.

- Gradient Descent
 - Stochastic Gradient Descent (SGD)
 - Adam
 - RMSProp
 - Adagrad
-

47. What is stochastic gradient descent (SGD)?

SGD updates weights using one training sample at a time.

48. Differentiate between batch gradient descent and stochastic gradient descent.

Batch Gradient Descent Stochastic Gradient Descent

Uses full dataset

Uses one sample

Slower

Faster

Batch Gradient Descent Stochastic Gradient Descent

Stable updates

Noisy updates

49. What is overfitting?

When the model learns training data too well and performs poorly on new data.

50. What is underfitting?

When the model cannot learn properly from training data and gives poor results.

51. Why is data preprocessing important?

It improves data quality and helps the neural network learn better and faster.

52. What is normalization?

Normalization converts data into a smaller common range, usually between 0 and 1.

53. What is feature scaling?

Feature scaling adjusts values of features so all features have similar ranges.

54. Why do we split data into training and testing datasets?

- Training data → Used to train the model
 - Testing data → Used to check model performance on new data
-

55. What is validation data?

Validation data is used during training to tune the model and prevent overfitting.

56. What is one-hot encoding?

A method used to convert categorical data into binary format (0 and 1).

Example:

Red = [1 0 0]

Blue = [0 1 0]

57. What type of data is suitable for ANN?

Numerical and properly preprocessed data is suitable for ANN.

58. What are categorical and numerical features?

- **Categorical features:** Text or labels (e.g., color, gender)
 - **Numerical features:** Numbers (e.g., age, salary)
-

59. Why is missing data problematic?

Missing data reduces model accuracy and can give incorrect predictions.

60. How can ANN handle noisy data?

By using:

- Data cleaning
 - Regularization
 - More training data
 - Proper preprocessing
-

61. Which libraries are commonly used for ANN implementation in Python?

- TensorFlow
 - Keras
 - PyTorch
 - NumPy
 - Pandas
-

62. What is TensorFlow?

TensorFlow is an open-source library used for machine learning and deep learning.

63. What is Keras?

Keras is a simple deep learning library used to build neural networks easily.

64. Differentiate between TensorFlow and Keras.

TensorFlow	Keras
Backend deep learning framework	High-level API
More complex	Easy to use
Powerful for large projects	Best for beginners

65. What is PyTorch?

PyTorch is an open-source deep learning framework mainly used in research and AI development.

66. How do you create a neural network model in Keras?

By using `Sequential()` and adding layers like Dense layers.

Example steps:

- Import Keras
 - Create model
 - Add layers
 - Compile model
 - Train model
-

67. What is the purpose of model compilation?

Compilation prepares the model for training by selecting optimizer, loss function, and metrics.

68. What parameters are specified during model compilation?

- Optimizer
 - Loss function
 - Metrics (like accuracy)
-

69. What is model fitting?

Model fitting means training the neural network using training data.

70. Explain the use of `Sequential()` model in Keras.

`Sequential()` is used to create neural networks layer by layer in a simple linear sequence.

71. What is the role of Dense layer?

Dense layer connects every neuron to every neuron in the next layer.

72. How do you evaluate a trained model?

Using testing data and checking metrics like accuracy and loss.

73. What is the use of predict() function?

predict() is used to generate output predictions from new input data.

74. What is accuracy in ANN?

Accuracy measures how many predictions are correct.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

75. How can you improve model accuracy?

- Use more data
- Tune hyperparameters
- Use better preprocessing
- Add hidden layers
- Reduce overfitting
- Choose proper activation functions

76. Why is differentiation used in backpropagation?

Differentiation helps calculate how much the weights should change to reduce error.

77. What is chain rule in ANN?

Chain rule is used in backpropagation to calculate gradients layer by layer from output to input.

78. Explain weight update rule.

Weights are updated to minimize error during training.

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

Where:

- η = learning rate
 - L = loss function
-

79. What is the role of derivatives in ANN learning?

Derivatives show the direction and amount of change needed to reduce error.

80. Why should activation functions be differentiable?

Because backpropagation requires derivatives to update weights correctly.

81. Explain the ANN experiment you performed.

We created and trained an ANN model using training data to perform prediction/classification and checked its accuracy.

82. What dataset did you use in your practical?

Example answer:

We used a dataset containing input features and output labels for training and testing the ANN model.

(You can replace this with your actual dataset name.)

83. What preprocessing steps were applied?

- Data cleaning
- Normalization
- Feature scaling
- Splitting data into training/testing sets

84. Which activation function did you use and why?

We used **ReLU** because it is fast and reduces vanishing gradient problems.

85. How many hidden layers did your model contain?

Example answer:

The model contained 1 or 2 hidden layers.

(You can change according to your project.)

86. Which optimizer did you use?

We used the **Adam optimizer** because it gives fast and efficient training.

87. What learning rate was selected?

Example answer:

The learning rate selected was 0.001.

88. How did you measure performance?

Using metrics like:

- Accuracy

- Loss
 - Confusion matrix
-

89. What accuracy did you achieve?

Example answer:

The model achieved around 90% to 95% accuracy.

(Replace with your actual result.)

90. What challenges did you face during implementation?

- Overfitting
 - Data preprocessing issues
 - Long training time
 - Parameter tuning
-

91. Mention real-life applications of ANN.

- Image recognition
 - Speech recognition
 - Medical diagnosis
 - Recommendation systems
 - Self-driving cars
-

92. How is ANN used in image recognition?

ANN learns image patterns and identifies objects, faces, or digits from images.

93. How is ANN useful in medical diagnosis?

It helps doctors detect diseases from medical data and images.

94. What is the role of ANN in speech recognition?

ANN converts spoken language into text by learning voice patterns.

95. How is ANN used in recommendation systems?

ANN recommends products, movies, or songs based on user preferences.

Example: Netflix recommendations.

96. Can ANN be used for weather prediction?

Yes, ANN can analyze weather data and predict temperature, rainfall, and climate conditions.

97. How is ANN applied in autonomous vehicles?

ANN helps self-driving cars detect roads, traffic signs, and obstacles.

98. What are the advantages of ANN?

- Learns complex patterns
 - High accuracy
 - Handles large data
 - Improves automatically through learning
-

99. What are the disadvantages of ANN?

- Requires large data
 - High training time
 - Needs powerful hardware
 - Difficult to interpret
-

100. Compare ANN with traditional programming approaches.

ANN	Traditional Programming
Learns from data	Uses fixed rules
Can improve accuracy	No self-learning
Handles complex problems	Best for simple logic
Adaptive system	Static system